Augmented Reality Experience for Car Modification

An AR Expereince Application on Smart Phone

CAR STYLER
AR/VR EXPERIENCE

Ayaan Ahmad

Problem Statement

Car owners struggle to visualize how styling stickers will look on their vehicles before applying them. This often leads to uncertainty and unsatisfying design choices.

Project Objective

To design a mobile-based AR application that allows users to visualize and experiment with styling stickers on their cars in real time using their phone's camera

Design Features

- Overlay stickers directly on the actual car.
- Scale, rotate, position and lock.
- Pre-designed car stickers gallery.
- Dimension Display
- Flashlight for better visibility in low light.
- Take a photo & save to gallery

Possible Design Solution →

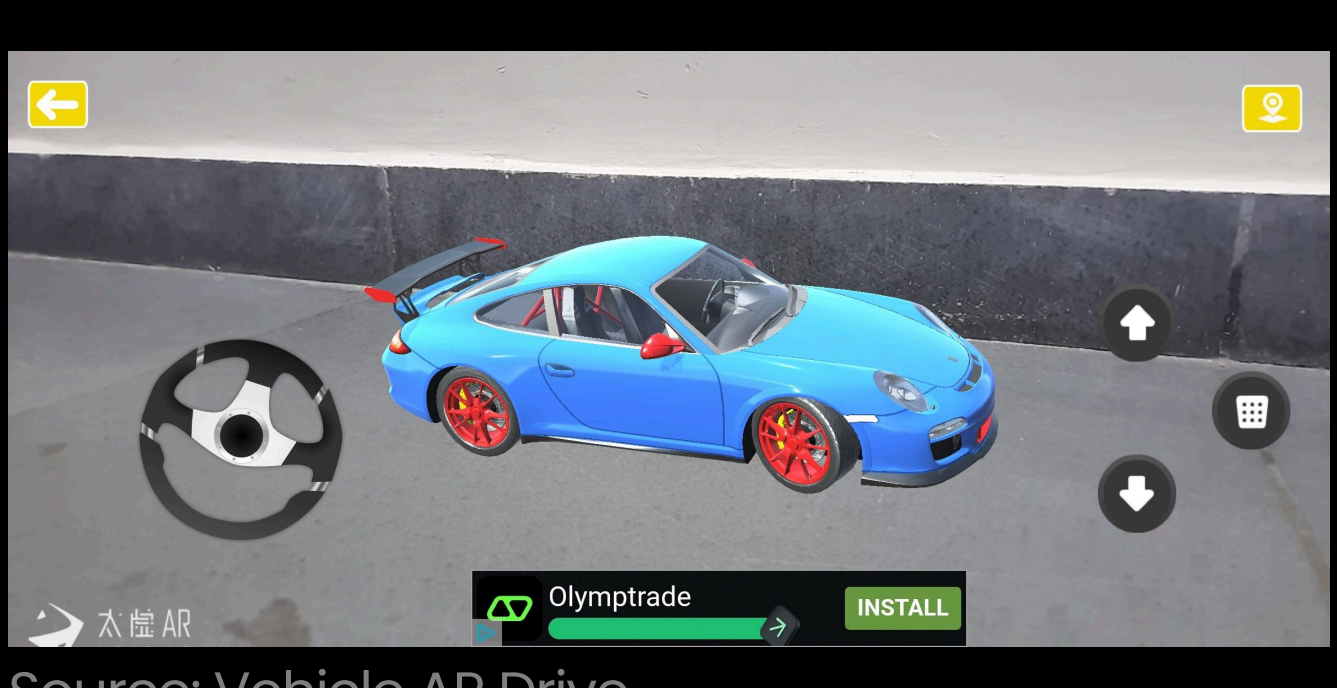
Online Research

Source: Vection Technologies

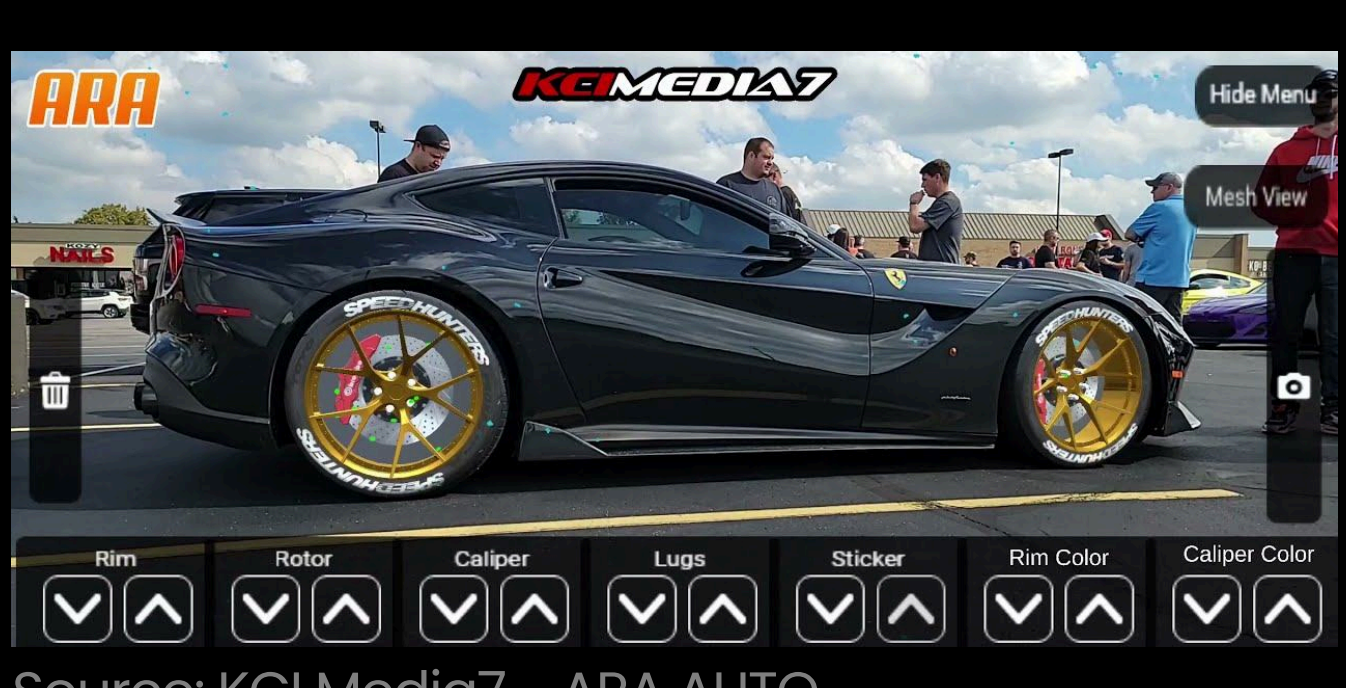
AR in the Automotive Industry: Trends & Use Cases (2025)

AREA	KEY USE CASES & EXAMPLES
Design & Manufacturing	Virtual prototyping, fitment checks (BMW, VW)
Displays & Navigation	AR HUDs, full-windshield displays (BMW, Hyundai, AllFocal)
Maintenance & Training	DIY guides, diagnostics overlays, assembly training
Marketing & Sales	AR configurators, virtual showrooms, immersive campaigns
Passenger Experience	In-cabin AR/VR entertainment and mixed-reality experiences

Inspiration Gathering



Source: Vehicle AR Drive



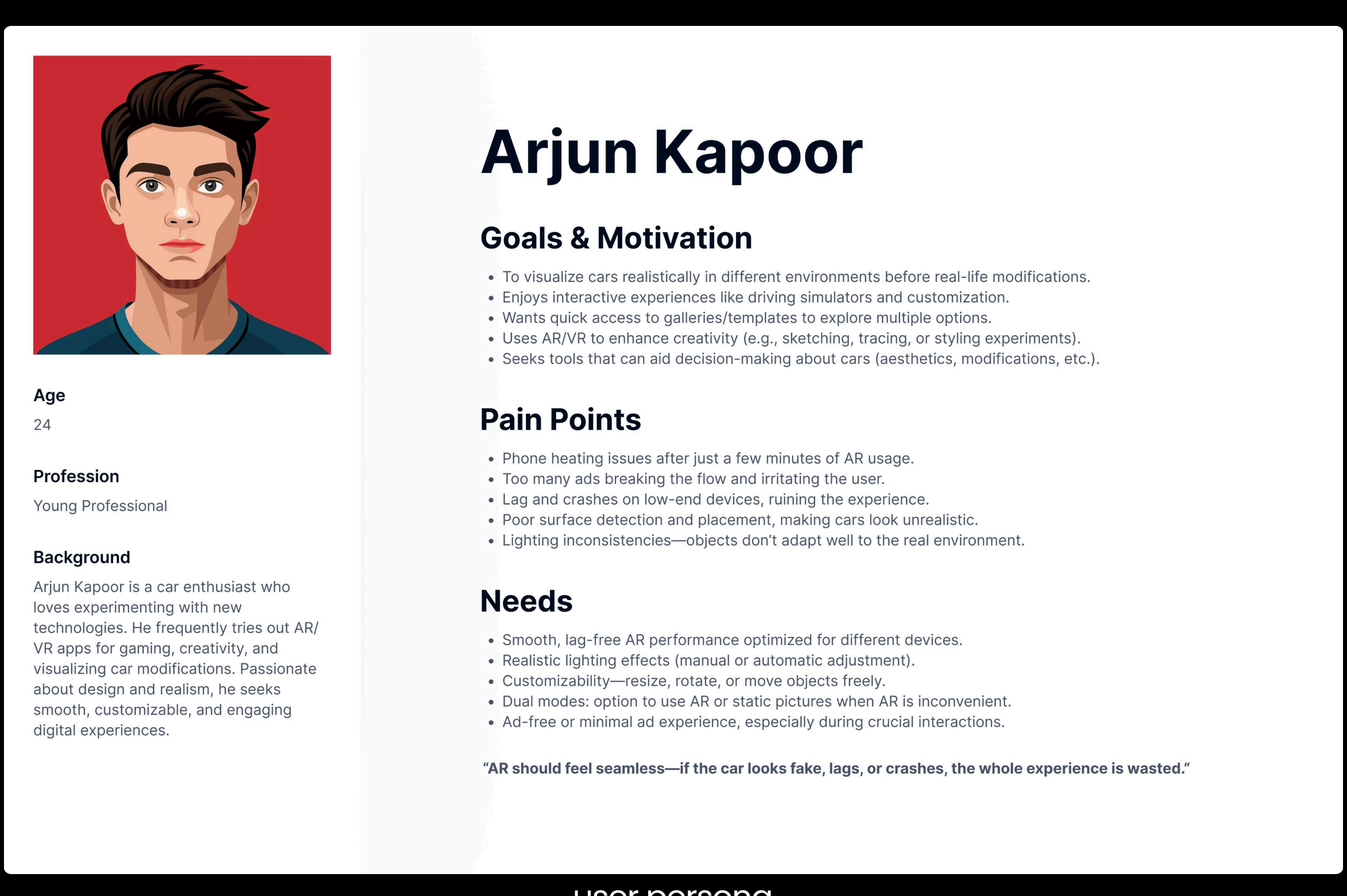
Source: KCI Media7 - ARA AUTO

Source: Porsche Newsroom

- I was inspired by features like resizing, public galleries, and real-time placement, and many more features which spark creativity and engagement.
- The limitations and problems that I noticed motivated me to design a smoother, more user-friendly AR app for 'CAR STYLING'.

Moodboard

Understanding the User Needs



Arjun Kapoor

Age
24

Profession
Young Professional

Background
Arjun Kapoor is a car enthusiast who loves experimenting with new technologies. He frequently tries out AR/VR apps for gaming, creativity, and visualizing car modifications. Passionate about design and realism, he seeks smooth, customizable, and engaging digital experiences.

Goals & Motivation

- To visualize cars realistically in different environments before real-life modifications.
- Enjoys interactive experiences like driving simulators and customization.
- Wants quick access to galleries/templates to explore multiple options.
- Uses AR/VR to enhance creativity (e.g., sketching, tracing, or styling experiments).
- Seeks tools that can aid decision-making about cars (aesthetics, modifications, etc.).

Pain Points

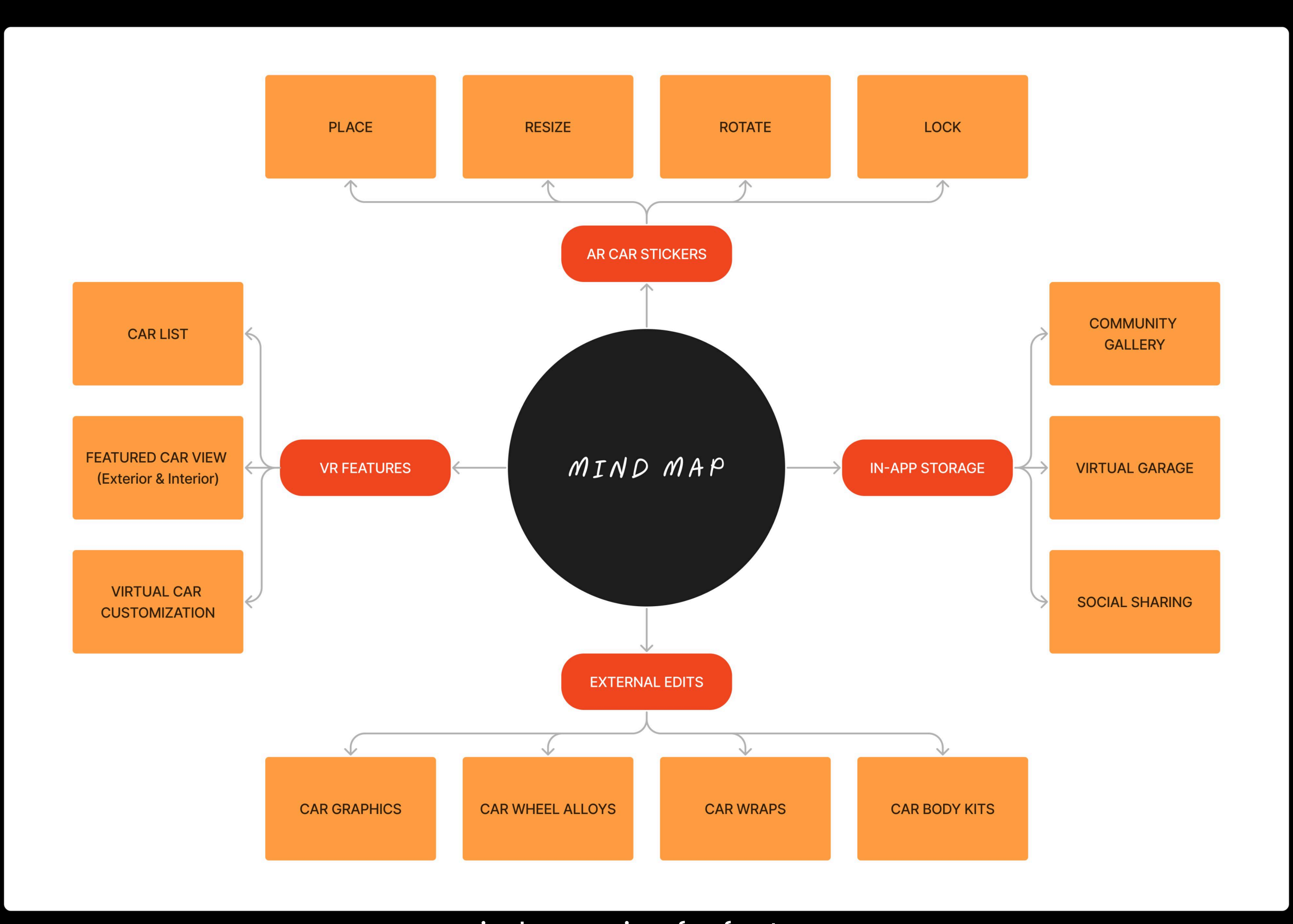
- Phone heating issues after just a few minutes of AR usage.
- Too many ads breaking the flow and irritating the user.
- Lag and crashes on low-end devices, ruining the experience.
- Poor surface detection and placement, making cars look unrealistic.
- Lighting inconsistencies—objects don't adapt well to the real environment.

Needs

- Smooth, lag-free AR performance optimized for different devices.
- Realistic lighting effects (manual or automatic adjustment).
- Customizability—resize, rotate, or move objects freely.
- Dual modes: option to use AR or static pictures when AR is inconvenient.
- Ad-free or minimal ad experience, especially during crucial interactions.

"AR should feel seamless—if the car looks fake, lags, or crashes, the whole experience is wasted."

user persona



mind mapping for features

STORYBOARDING

"FROM DISSATISFACTION TO ASSURANCE" - ARJUN'S EXPERIENCE

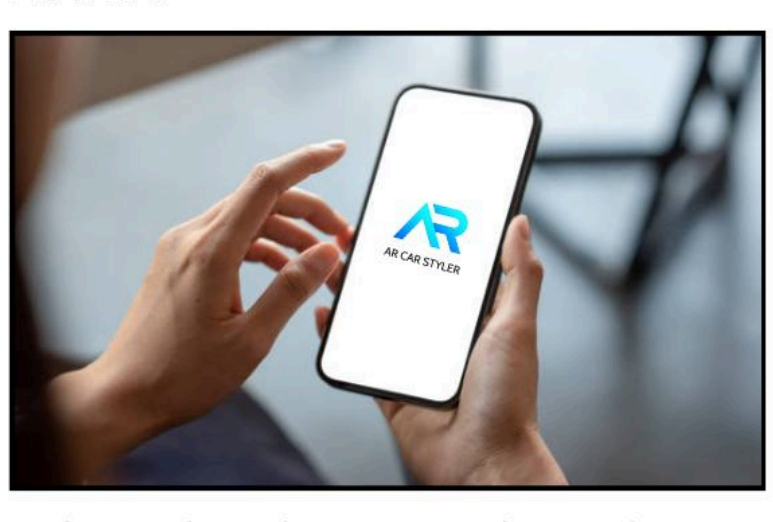
FRAME 1

Arjun went to a car modification store to get his car graphics done, but he was dissatisfied with the results.

FRAME 2

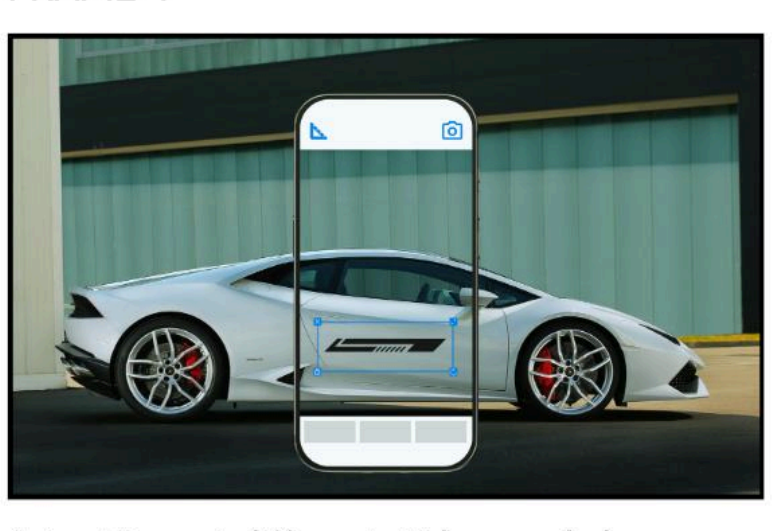
He manages to repair the defective sticker, but he still needs graphics for his car.

FRAME 3



He learns about the AR Car Styler App from one of his friends, who keeps up with car detailing.

FRAME 4



Arjun tries out different stickers and places each one on his car to assist him make style selections before putting physical stickers.

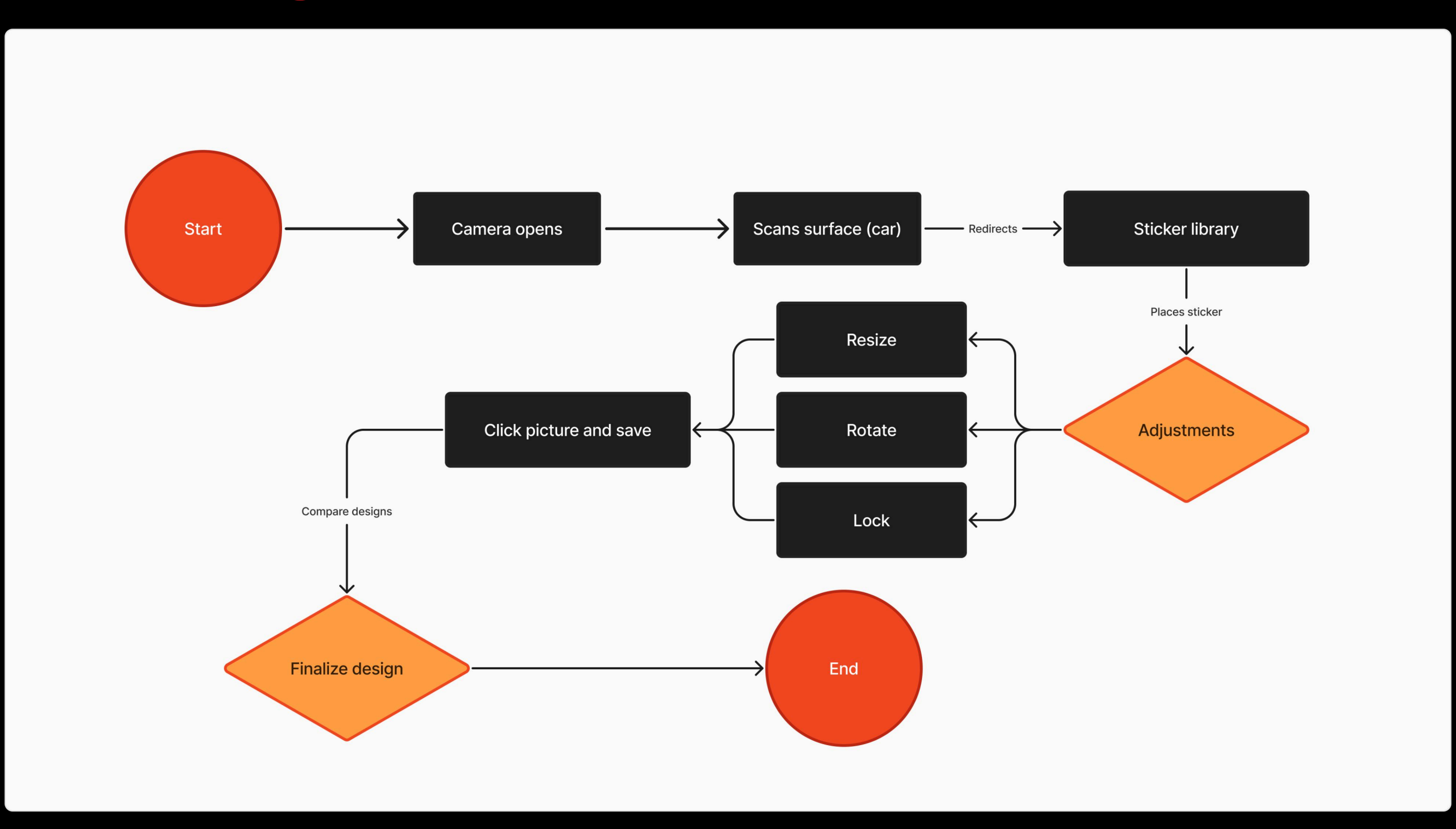
FRAME 5

He finalizes a sticker and saves the image to his phone so that it can be applied on his car in real life.

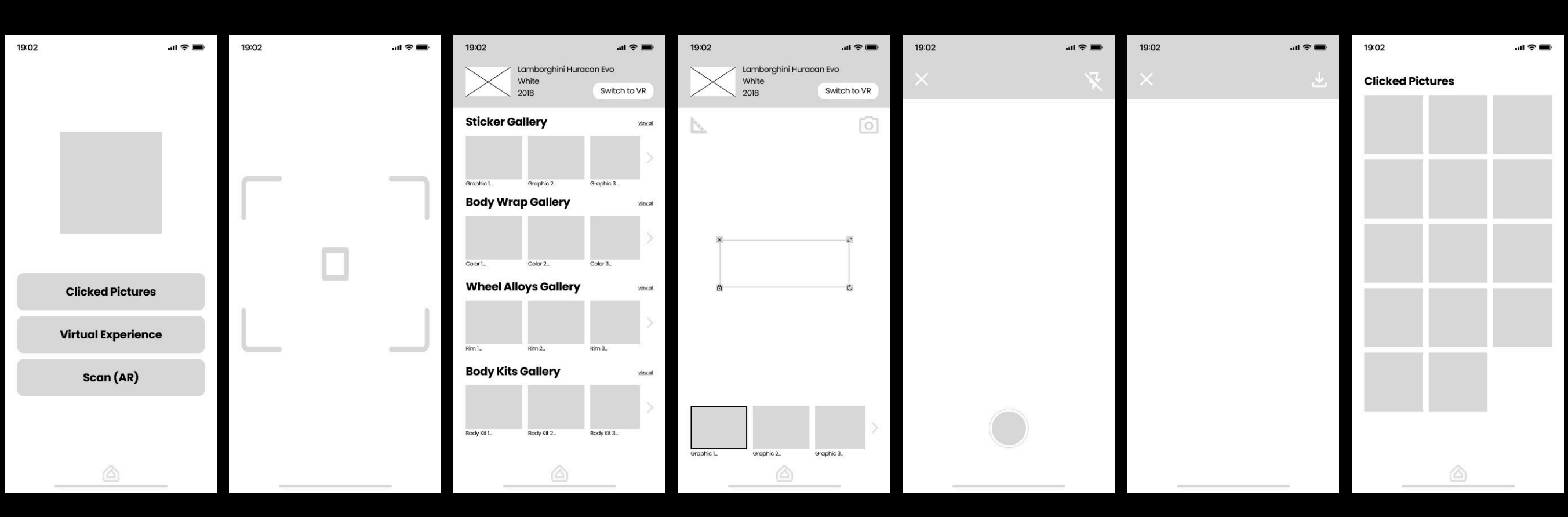
FRAME 6

He finally receives the graphics of his selection and is really satisfied with the outcome.

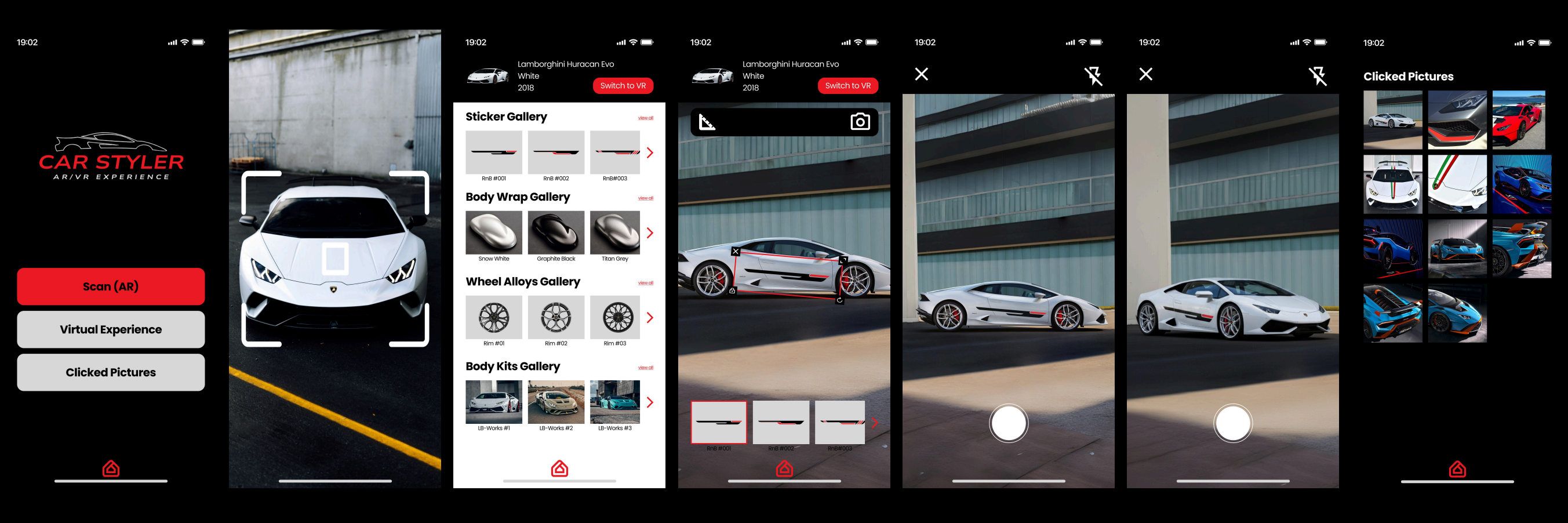
Creating a User Flow for Screens



Wireframes



High Fidelity UI



FIGMA PROTOTYPE

FIGMA FILE

AR CAR STYLER COMPILATION VIDEO

Ayaan Ahmad